

$$P(Token_k = w \mid Token_{<k}) = \frac{\exp(\text{Proj}(\mathbf{a}_k)_w)}{\sum_{w' \in \mathcal{V}} \exp(\text{Proj}(\mathbf{a}_k)_{w'})}, \quad \text{NLL}(Token_k = w) = -\log \frac{\exp(\text{Proj}(\mathbf{a}_k)_w)}{\sum_{w' \in \mathcal{V}} \exp(\text{Proj}(\mathbf{a}_k)_{w'})}$$